

Experiment No: 14

Title

Write a Python Program to Implement Alpha-Beta Pruning Algorithm for Gaming

Aim

To develop a Python program to implement the **Alpha-Beta Pruning** algorithm for finding the optimal move in a game.

Objective

To implement the Alpha-Beta Pruning algorithm for reducing unnecessary node evaluation while determining the optimal game value.

Algorithm

Step 1: Start the program.

Step 2: Define the leaf node values.

Step 3: Initialize alpha as $-\infty$ and beta as $+\infty$.

Step 4: Traverse the game tree recursively.

Step 5: If it is the maximizing player's turn, choose the maximum value and update alpha.

Step 6: If it is the minimizing player's turn, choose the minimum value and update beta.

Step 7: If **alpha $\geq$ beta**, prune the remaining branches.

Step 8: Display the optimal value.

Step 9: Stop the program.

Program

```
# Alpha-Beta Pruning Algorithm
```

```
scores = [3, 5, 2, 9, 12, 5, 23, 23]
```

```
def alphabeta(depth, node, alpha, beta, maximizing):
```

```
if depth == 3:
    return scores[node]

if maximizing:

    value = float('-inf')

    for i in range(2):

        value = max(value,
                    alphabeta(depth + 1,
                               node * 2 + i,
                               alpha,
                               beta,
                               False))

    alpha = max(alpha, value)

    if alpha >= beta:
        break

    return value

else:

    value = float('inf')

    for i in range(2):
```

```

        value = min(value,
                    alphabeta(depth + 1,
                              node * 2 + i,
                              alpha,
                              beta,
                              True))

    beta = min(beta, value)

    if alpha >= beta:
        break

return value

result = alphabeta(0, 0, float('-inf'), float('inf'), True)

print("Optimal Value :", result)

```

Sample Input

No user input.

Sample Output

```

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Optimal Value : 12
|

```

Result

Thus, the Python program to implement the **Alpha-Beta Pruning Algorithm** was executed successfully, and the optimal game value was obtained by pruning unnecessary branches.

